

COMPARISON OF VARIOUS RF COMMUNICATION TECHNOLOGIES

Technical capabilities	Low power wide area networks (LPWAN)							Short range networks				
	LoRaWAN	Nb-IoT	Nb-IoT	Sigfox	Weightless-N	Weightless-P	Cellular	BLE	WiFi	Thread	ZigBee	Z-Wave
Range	2-5km urban; 15km suburban; 45km rural	Up to 10km	Up to 10km	Up to 10km urban; 50km rural	5km	2km	35km GSM; 200km 3G/4G	80m	50m	Mesh	100m Mesh	30m Mesh
Deep indoor performance	Yes	ISM yes, whitespace no	Yes	Yes	Yes	Yes	No	No	No	No	-	-
Freq. band	Varies, Sub-GHz	ISM or whitespace	Sub-GHz	Frequency-independent, 868/902MHz	Sub-GHz	Sub-GHz	900/1800/1900/2100MHz	2.4GHz	2.4GHz	2.4GHz	915MHz/2.4GHz	900MHz
ISM	Yes	Yes, depends on base-station	Yes	Yes	Yes	Yes	Depends	Yes	Yes	Yes	Yes	Yes
Fully bi-directional	Yes, depends on mode	Yes	No	No	Uplink only	Yes	Yes	Yes	Yes	-	Yes	Yes
Data rate	0.3-50kbps adaptive	10-100kbps	100bps	10-1000bps	30kbps-100kbps	Up to 100kbps adaptive	35-170kbps GSM/3-10Mbps LTE	< 1mbps	600Mbps max	-	250kbps	10-100kbps
Power profile	Low	Low	Low	Low	Low	Low	Medium	High	High	Low	Low	Low
Authentication	Yes	-	Yes	Yes	Yes	Yes	High security, backed by major telecoms	Trusted devices problematic	Yes	Yes	Yes	Yes
E2E encryption	Yes	-	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Over-the-air software upgrades	Yes	-	No	No	No	Yes	Yes	Yes	Yes	-	Yes	Yes
Supports sensors moving between hubs	Yes	-	No	No	Yes	Yes	Yes	Yes	Yes	No	Yes, mesh-based	Yes, mesh-based
Location aware	Yes	-	No	No	No	-	Yes	No	Yes	-	-	-
Operational model	Public or private	-	Public or private	Public	Public or private	Public or private	Public or private	Public or private	Public or private	Private/ WiFi backbone	Public or private	Public or private
Standard	LoRaWAN	Weightless	Weightless	No	Weightless	Weightless	GSM, LTE etc	Bluetooth 4.0	IEEE 802.11	Thread, based on 802.15.4	ZigBee	Z-wave
Scalability	Yes	Yes	Yes	Yes	Limited	Yes	Yes	Yes	-	Yes	Yes	Limited

(Source: <https://s3-eu-west-1.amazonaws.com>)



Fig. 4: IR transmitter (Image source: www.robotshop.com)

and hospitals to communicate patient data (as light travels faster).

Infrared (IR). Personal communication is best enjoyed with IR. Electromagnetic invisible light range allows data communication over a short distance. Devices used as the medium are personal computers and digital assistants.

Bluetooth

Wireless headphones connect to your cellphone using Bluetooth technology. Assigned 802.15.1 standard by IEEE,

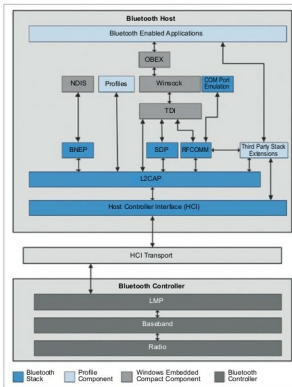


Fig. 5: Bluetooth architecture (Source: <https://msdn.sec.s-msft.com>)

Bluetooth is used in various personal and commercial devices. The fact that it can be used for small products like wireless headphones and smart watches makes it a favourite among tech-savvy individuals. Bluetooth Low Energy, a Bluetooth variant launched in 2011, consumes less power.

The road ahead

The IoT wave is unstoppable. At the same time, mass adoption of smart living calls for foolproof security. Informed consent for data sharing is important to ensure safety of consumer data. **RFY**